MILITARY SPECIFICATION

GEAR BEVEL, INSPECTION OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the methods of measurement and requirements for acceptance of the fundamental errors for straight, zerol, and spiral bevel gears when the control is based upon the use of master gears. The manufacture of bevel gears, particularly precision and those requiring utmost quietness in operation, depends vitally on a careful inspection procedure. Because of the complex configurations in bevel gears and the need for conjugate tooth action, it is necessary that diligent inspection be carried out through the whole manufacturing process. In addition, this specification covers that portion of the developmental operation which directly affects assembly and inspection procedures.

1.2 Classification. Bevel gears shall be one of the types shown in table I.

TABLE I. Bevel gear types.

Item	Bevel		Hypoid	
Gear type:	Straight	Coniflex	Spiral	Zerol
Tooth form:	Octoid	Spherical	Revacycle	Involute
	Formate	Helixform	Duplex helical	

2. APPLICABLE DOCUMENTS

2.1 This section is not applicable to this specification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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3. REQUIREMENTS

3.1 Development gears. The first step in the manufacture of bevel gears is to produce a trial or development pair of gears which pass inspection and are approved for operation in the final mounting. Following approval several sets of master control gears are produced which duplicate the "developed pair." These hardened control gears are used in the testing machines to control quality at all stages of manufacture.

3.1.1 Straight (coniflex type), zerol and spiral bevel gears.

3.1.1.1 Determining optimum tooth bearing. The development procedure for these gears shall include determination of the optimum tooth bearing with both members of a pair positioned at their normal operating distances and zero offset and run together under a light brake load. Values for normal operating distances are given in the bevel gear data of each level gear drawing. The optimum tooth bearing for the above types of gears is one meeting the following requirements.

3.1.1.1.1 Bearing area location. The bearing area shall be slightly nearer to the toe than to the heel and shall be centrally located between the top land and start of fillet. Any deviation from this central location shall be slightly high on the pinion and slightly low on the gear. Figure 1 shows the type of bearing desired.

3.1.1.1.2 Bearing area size. The size of the bearing area shall be large as possible without causing it to extend to any of the tooth edges for any extreme combinations of operating conditions. These combinations shall be taken in such a manner as to include vertical offset variations and the operating distance variations for both members.

3.1.1.1.3 Bearing area shape. The bearing area shall be essentially of the shape shown in figure 1. It shall have a smooth outline with no abrupt changes and shall be uniform from tooth to tooth. In certain specific cases, it will be necessary to define the optimum pattern more thoroughly. The required pattern shall be in accordance with the requirements of table II.

3.1.1.1.4 Extent of bearing area. The tooth bearing area shall not extend to any of the tooth edges at any of the extreme combinations of the following variables taken together:

(a) Operating distance of master gear
(b) Operating distance of mating master gear
(c) Vertical offset

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Vertical offset does not refer to vertical with respect to the orientation of the gears in their final mountings. It refers only to the sense of movement in the testing machine. In addition, the value given on each bevel gear drawing is the total value for the set rather than an individual figure for one gear.

3.1.1.1.5 Establishing and approving tooth bearing patterns. The preceding requirements represent the optimum tooth bearing patterns developed under ideal conditions. Various factors such as gear tooth geometry, operating conditions and methods of manufacture may make it necessary to depart from these requirements. The cognizant engineering activity shall be responsible for establishing approval of these tooth bearing patterns.

3.1.1.2 Developed gear set noise. The developed gear set shall run together smoothly without unusual or excessive noise when checked in the bevel gear testing machine.

3.1.1.3 Drawing limits. The developed gear set shall conform to drawing limits for backlash, runout, tooth spacing and location of top land and back face.

3.1.1.4 Tooth bearing diagrams. Once having established the optimum tooth bearing, it is necessary to run the set at their respective reference mounting distances and to observe the resulting tooth bearing. A diagram shall be made of both patterns, namely:

(a) With both members at their respective normal operating distances and zero offset.
(b) With both members at their respective reference mounting distances and zero offset.

These diagrams shall be approved by the cognizant engineering activity. One set of diagrams shall be included in the process sheets for each gear. In addition, inspection and assembly shall each be furnished duplicate copies of the diagrams.

3.1.2 Straight bevel gears not of the coniflex type.

3.1.2.1 Development procedure. The development procedure for these gears shall include determination of the optimum tooth bearing with both members at their respective reference mounting distances and zero offset. Values for reference mounting distances are given on every bevel gear drawing. The optimum distances are given on every bevel gear drawing. The optimum tooth bearing for the above gear type is defined to be one meeting the following requirements.

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3.1.2.1.1 Bearing area. The bearing area shall be substantially full length. It should begin at the small end of the tooth and extend about three-quarters length of the tooth. Figure 2 gives the desired bearing size and location. The bearing is higher on the pinion than on the gear tooth. The development of these gears is to the fixed reference mounting distances and no variations in mounting conditions need be considered.

3.1.2.1.2 Shape. The bearing area shall be essentially of the shape shown in figure 2. It shall have a smooth outline with no abrupt changes and shall be uniform from tooth to tooth.

3.1.2.1.3 Pattern approval. The optimum pattern established for each set of these gears shall be approved by the cognizant engineering activity.

3.1.2.2 Developed gear set. The developed gear set shall run together smoothly and without unusual or excessive noise, when checked in the bevel gear testing machine.

3.1.2.3 Check drawing requirements. The developed gear set shall conform to drawing limits for backlash, runout, tooth spacing, and location of top land and back face.

3.1.2.4 Prepare diagram. Once having established the optimum tooth bearing with both members at their respective reference mounting distances, a diagram shall be made of that pattern. One diagram shall be included in the process sheets for each gear. In addition, inspection and assembly shall each be furnished duplicate copies of the diagram.

3.2 Production gears.

3.2.1 Tooth bearing test for straight (coniflex type), zerol and spiral bevel gears.

3.2.1.1 Development procedure. The development procedures establish the desired tooth bearing for the following two mounting conditions.

(a) Both members at normal operating distance and zero offset.
(b) Both members at reference mounting distance and zero offset.

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3.2.1.2 Check tooth bearing patterns. It is required that each gear have its tooth bearing checked under these mounting conditions when meshing with its mating production control gear. The bearing patterns obtained shall closely duplicate the established standards of acceptability for production control gears (see figure 3).

3.2.1.3 Bearing pattern uniformity. It is required that the tooth bearing be uniform from tooth to tooth. If there is non-uniformity of such a degree to make acceptance or rejection questionable, it is required that both the runout and the tooth spacing be checked. Should these two values be within drawing limits, the gear may be accepted.

3.2.1.4 Running noise. Excessive running noise or rough action during the bearing test shall be considered sufficient cause for rejection of the gear. When the noise or roughness is such that acceptance or rejection would be questionable it is required that both runout and tooth spacing be checked. Should these two values be within drawing limits the gear may be accepted.

3.2.2 Tooth bearing test for straight bevel gears not of the coniflex type.

3.2.2.1 Development procedure. The development procedure establishes the desired tooth bearing with both members at their respective reference mounting distances and zero offset. It is required that each production gear have its tooth bearing checked under these same mounting conditions when meshing with its mating production control gear. The bearings shall closely duplicate those established in development. It is not required that these gears be checked at any combination of mounting variations other than the one combination specified, namely:

(a) Zero vertical offset.
(b) Both production gear and mating production control gear at the respective reference mounting distances for the production set.

3.2.2.2 Tooth bearing uniformity. It is required that the tooth bearing be uniform from tooth to tooth. If there is non-uniformity to make acceptance or rejection questionable, it is required that both the runout and tooth spacing be checked. Should these two values be within drawing limits, the gear may be accepted.

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3.2.2.3 Excessive running noise. Excessive running noise or rough action during the bearing test shall be considered sufficient cause for rejection of the gear. When the noise or roughness is such that the acceptance or rejection would be questionable, it is required that both runout and tooth spacing be checked. Should these two values be within drawing limits the gear may be accepted.

3.2.3 Other. The acceptable limits for the following characteristics are given on the drawing for each specific bevel gear.

(a) Location of top land and back face.
(b) Backlash.
(c) Runout.
(d) Tooth spacing.

NOTE: Because of the length of time required to inspect a gear for runout or tooth spacing, they are only required when any of the following is detected in the gear tests:

(1) Non-uniform tooth bearing from tooth to tooth.
(2) Questionable noise.
(3) Questionable degree of roughness in operation.

3.2.4 Common gear. When one gear mates with two or more gears at the same time and two or more meshes have a different developed tooth bearing pattern, the common gear shall have the proper developed tooth bearing pattern with all its mating production control gears. The common gear need only be checked with one of its mating production control gears to save time, but shall conform to the above requirement.

3.3 Master control gear.

3.3.1 Model master control gears. The model master control gears shall conform to the same requirements as the production gear specified in 3.2, with the following exceptions.

3.3.1.1 Tooth bearing test. The model master control gear shall be run with its mating model master control gear in the tooth bearing test.

3.3.1.2 Marking master control gears. The tooth thickness of the model master shall be made equal to the theoretical tooth size minus a portion of whatever backlash manufacturing may deem necessary or suitable to them. Usually one-half of the total required backlash (with both model masters at minimum operating distance and zero offset)

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will be cut into each of the gears. The amount cut into each gear shall be clearly and permanently etched on the model master gear to the nearest ten thousandth of an inch. It is emphasized that this etched value is not the total backlash for the set of model masters, but is an individual figure for the one member. The total backlash for the set (with both model masters at minimum operating distance and zero offset) is therefore equal to the sum of the etched values for the two model masters.

3.3.1.3 Runout. The runout of any master control gear shall not exceed 0.0005 inch.

3.3.1.4 Use of master control gears. Master control gears are used only to check the inspection control gears.

3.3.2 Inspection control gears.

3.3.2.1 Inspection control gears. Inspection control gears shall conform to the same requirements as the master control gear specified in 3.3.

3.3.2.2 Check for wear. Inspection control gears shall be checked periodically for wear. Should the total backlash for the set (with both members at minimum operating distance and zero offset) exceed the sum of the etched values by 0.0005 inch or more, the etched values shall be corrected accordingly.

3.3.3 Production control gears. The production control gears shall conform to the same requirements as the production gear specified in 3.2, with the following exceptions:

3.3.3.1 Tooth bearing test. The basic production control gear shall be run with the mating production control gear in the tooth bearing test.

3.3.3.2 Backlash. The tooth thickness of the production control gear shall be controlled by limiting its backlash when in mesh with its mating inspection control gear (with both members at their respective minimum operating distance and zero offset). This backlash shall be B + .001, -.000 where B is that portion of the total backlash between the set of inspection control gears that have been etched on the mating inspection control gear.

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3.3.3.3 Production control gear. Each production control gear shall be periodically checked for wear. This shall be done by measuring the backlash when in mesh with its mating inspection control gear (both members at their respective minimum operating distances and zero offset). Should the measured backlash ever exceed the value B + .001, where B is the value etched on the inspection control gear, the inspection control gear shall be inspected to determine whether the excess backlash is due to any permissible wear of the inspection control gear. If the value determined by the inspection of the inspection control gear coincides with the etched value, the production control gear is no longer acceptable for use. Should the inspection control gear show any amount of permissible wear, both the upper and lower limits of the backlash between the production and the inspection control gear (with both members at their respective minimum operating distances and zero offset) shall be increased by an amount equal to the amount of wear of the inspection control gear. If the backlash between the production control gear and mating inspection control gear falls within this new range of backlash, then the production control gear is acceptable for use. If the backlash between the production control and inspection control gear does not fall within this corrected range, the production control gear is not acceptable for use.

3.3.3.4 Runout. The runout of any basic master gear shall not exceed 0.0005 inch.

3.3.4 Mandatory. When one gear mates with two or more gears at the same time and two or more meshes have a different developed tooth bearing pattern, control gears shall be established for all gears.

3.3.5 Hardening. All gears that have been accepted as meeting the requirements of this specification and have been run in with the green production control test shall be surface hardened in accordance with the bevel gear drawing requirements.

3.3.6 Final test. A final test is usually made on each pair after hardening, or in the case of curved-tooth bevel gears, after subsequent lapping or grinding. Where adjustability is provided in the final mounting, the testing machine provides a rapid and easy means of determining, for each pair of gears, the mounting distances at which they operate best. In this case, the gears are matched in sets and usually wired together. It is common to mark the measured "mounting distance" and "backlash" values on the outer end of the teeth.

3.3.7 Finish. Surface finish of the gears shall be as specified on applicable part drawings. Gears shall be deburred and sharp edges removed.

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3.3.8 Marking. Each gear shall have identification markings etched or stamped as specified on applicable drawings, or in accordance with contract requirements. Each bevel gear set shall be serialized and permanently marked as matched sets, and shall be stocked and issued as matched sets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies shall be cause for suspension of acceptance until correction has been made and conformance of a product to specification criteria has been demonstrated.

4.3 Quality conformance inspection.

4.3.1 100% inspection. Each bevel gear manufactured shall be matched with the mating gear or pinion and mounted on a bevel gear testing machine provided with axial adjustments for the gear and pinion, and with a vertical offset adjustment for raising or lowering the axis of one member vertically with respect to the other member for the testing of hypoid gears. The following requirements shall be met.

(a) Tooth bearing pattern.
(b) Noise and smoothness of operation.
(c) Location of top land and back face.
(d) Backlash.
(e) Runout.
(f) Tooth spacing.

Gear sets meeting the above requirements shall be serialized and retained as matched sets.

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4.3.2 Failure. Gear sets not meeting the test requirements in 4.3 shall be rejected.

4.4 Test methods.

4.4.1 Tooth bearings. The test for bevel gears is a check of their operation under normal running conditions in their final mountings. Bevel gear testing machines are designed to provide a running test of the gear under simulated service conditions of running. Controlled axial and vertical movements are provided in the machine in order to simulate service deflections as well as to allow for the mis-positioning of the gears due to housing tolerances, bearing play, etc. In making this test, the production gear and its mating gear are mounted at their respective selected operating distances and run together under a light brake load. Prior to running, the gears are painted with a non-abrasive marking compound. A run of a few seconds is sufficient to rub off the compound in the area of tooth contact. The tooth contact pattern shall meet the standards shown in figures 1, 2 or 3 as applicable.

4.4.2 Noise and smoothness of operation. The test for noise and smoothness is a check of gear operation under normal running conditions in the final mountings. This test should be combined with the tooth bearing check described above. While the gears are running in mesh for the tooth bearing test, the operator should listen for any unusual noise types and levels. Any other indications of rough action should also be recorded.

4.4.3 Location of top land and back face. These surfaces may be checked as shown in either figure 4 (part A or B). However, inspection may develop other methods (for example, optical comparator methods) so long as they are adequate to insure conformance of the gear to drawing requirements. The drawing for each particular gear furnishes sufficient basic dimensions to properly position and zero the dial indicators (see figure 4, part A) or to properly position the contact points (see figure 4, part B). Figure 5 shows the specific basic dimensions given on the drawing. It should be noted that the basic diameter given is a dimension which is concentric with the X diameter. It is important that the set-up of the checking device be made to satisfy this requirement. Once having accomplished this set up, the method of figure 4 part A, checks the surface locations by direct reading of the dial indicators, while the method of figure 4 part B, makes use of a feeler gage to be inserted between the contact points and the production gear. For either method it is not necessary that surfaces be checked along their entire length. Two points reasonably close to either end (lengthwise) of the particular surface will be sufficient.

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NOTE: Diameter X will be specified on the drawing. It will be selected as that diameter upon which the gear is mounted in final assembly.

4.4.4 Backlash. The measurement of backlash is made in the gear testing machine. The gear being inspected is assembled in mesh with its mating gear. With the gear fixed, the rotation of the production gear is measured using a dial indicator as shown in figure 6. This measurement is taken normal to the active tooth surface at the extreme heel of the gear. This is a suggested method, other methods may be employed so long as they are equivalent to this. Care should be taken to convert all readings to normal backlash should the method employed not indicate this value.

4.4.5 Runout. On bevel gears the runout tolerance includes both axial and radial runouts (see figure 11). Measuring methods are defined in the following four (4) paragraphs.

4.4.5.1 Rolling check. Axial runout is measured in a bevel gear testing machine. The gear being inspected is rolled in tight mesh with a master control gear having a maximum runout of .0005 inches. The variation in the mounting distance depicted in figure 9 is the axial runout.

4.4.5.2 Runout bearing variation. Variations in runout may be observed by running the gears in a suitable test machine. The runout is characterized by periodic variation in sound during each revolution, and by the tooth bearing shifting progressively around the gear from heel to toe, and toe to heel. Figure 10 depicts these variations. This test also includes the effects of tooth element variations. Runout is observed by visually checking variation in contact pattern. The actual amount should be determined by either the single or double probe method defined in the following paragraphs.

4.4.5.3 Single probe runout check. Measurements made with a single probe as depicted in figure 11 are used to determine true runout variations when measured perpendicular to the surface of revolution. The surface of revolution of a bevel gear is the pitch cone. A single probe can also be used to isolate axial or radial effects on total runout tolerance.

4.4.5.4 Two probe runout check. As shown in figure 12, this device has two contact points 180 degrees apart, one a fixed stop, the other actuating a dial indicator reading to ten thousandths of an inch. The gear to be checked is mounted on a dead center, held to the vise,

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and indicator readings are taken on diametrically opposite teeth, with the contacts acting on the corresponding sides of these teeth. The contact points are mounted on a swinging bracket, which can be lifted while the gear is turned from tooth-to-tooth. The change between the largest and smallest readings obtained in one revolution of the gear is a value which is four times the eccentricity (twice the runout) of the pitch circle defined by the sides of the teeth. Thus, this test is a very sensitive test of the eccentricity of a gear. It measures eccentricity by a factor of 4, and tooth spacing errors by a factor of 1, thus the test has only minor uncertainty due to side effects.

4.4.6 Tooth spacing. The spacing of individual teeth of a bevel gear is measured in a Gleason tooth spacing tester or equivalent. This device has two contact points which can be adjusted to contact corresponding sides of adjacent teeth. One contact point is used as a fixed stop, and the other actuates a dial indicator reading to ten-thousandths inch. The two contact points are carried on a swinging bracket which can be lifted out of position and then returned to permit the checking of successive teeth in the gear. The indicator is set to zero for the first pair of teeth. All subsequent readings are therefore relative to the first pair of teeth selected. In interpreting the readings, it is important to note that the readings are not the tooth to tooth spacing errors. These errors are the differences between successive readings. The reliability of this check is limited by the sensitivity of the equipment and the skill of the operator. However, experience shows that the slight misinterpretations of the dial indicator readings are as often in the minus direction as they are in the plus direction, thereby cancelling each other when a reasonable number of teeth are checked. To illustrate the method, the readings for a ten tooth gear are given in table III.

4.4.7 Hardness check. Hardness checks shall be made to determine conformance to drawing requirements. A minimum of three (3) samples shall be selected from each heat treat lot for determination of effective case depth, core hardness and surface hardness. Samples shall be processed by the same heat treat process procedure used for production parts. Inspection and hardness tests shall be performed in the areas shown in figure 7.

4.4.7.1 Failure. Failure of any one (1) test sample shall be cause for rejection of the entire representative production lot. Items produced subsequent to failure of the test sample shall not be submitted for acceptance until corrective action has eliminated the cause of the failure. Acceptance of heat treat samples subsequent to failure shall not resume until three (3) consecutive samples have successfully met specified test requirements.

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4.4.8 Finish. Verify that the surface finishes specified on applicable part drawings conform to the specified requirements, and that there are no burrs or sharp edges on any part of the gears.

4.4.9 Marking. Verify that the gears are permanently marked as specified on applicable part drawings and that the bevel gear sets are serialized and maintained as matched sets.

5. PACKAGING

5.1 Packaging. Bevel gear sets shall be preserved, packaged, and packed in accordance with best commercial practice, or as specified in the contract or purchase order. Bevel gear set serial numbers shall be checked to assure packaging as matched sets.

6. NOTES

6.1 Intended use. The procedures covered by this specification are intended to be used to insure bevel gears meet prescribed inspection requirements.

6.2 Ordering data. Acquisition documents should specify title, number, and date of this specification. Bevel gears shall be manufactured, inspected, packaged or assembled, stocked, replaced, permanently identified and retained as serialized matched sets only.

6.3 Definitions.

6.3.1 Bevel gear. A bevel gear is essentially a truncated cone with a number of cogs or teeth fastened to the cone surface. Two bevel gears, when rotated in respective non-parallel, intersecting axes should transmit uniform rotary motion. While in mesh, the gears roll on cones called pitch cones. The apex of this cone for each of the two mating gears, is at the point of intersection of the axes.

6.3.2 Production gear. The gear which is being inspected.

6.3.3 Dimensional. Figure 8 gives most of the dimensional nomenclature pertinent to this specification. Other dimensional data is defined as follows:

(a) Reference mounting distance. This is the distance from the axis of one bevel gear to the thrust face of its mate as assembled in final mountings under no load. The final assembly is held to this dimension within the limits of accuracy imposed by shim lamination thickness. Original

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design is accomplished to insure that there will be sufficient shim stock for the extremes of tolerance build-up in the mountings.

(b) Operating distance. This is the actual distance, in final mountings from the axis of one bevel gear to the thrust face of its mate, for a specific operating condition. Three such operating distances may be specified on a bevel gear drawing. These are:

(1) Normal operating distance.
(2) Minimum operating distance.
(3) Maximum operating distance.

The normal operating position will usually be that at which the greatest amount of running time will be accumulated in services. The minimum and maximum values are based upon the deflections inherent in the given assembly under the expected range of loads.

(c) Theoretical tooth thickness. The thickness of tooth (measured at the large end) selected on the basis of a strength balance for the two mating gears of a set is the theoretical tooth thickness. These values are for zero backlash with the gears gears positioned at their respective reference mounting distances.

(d) Basic value. The value to which a specified tolerance is applied is the basic value. For example, if a dimension is given as 1.490 + .XXX, - .XXX inches, the basic value is 1.490 inches.

6.3.4 Bevel control gear. A bevel gear is made to the necessary specification of size, mounting distance, tooth element accuracy and tooth bearing to serve as a working standard for the duplication of acceptable and interchangeable gears in production. This specification requires the use of the following types of master gears:

(a) Master control gear. A bevel master gear used only for checking inspection control gears.

(b) Inspection control gear. A bevel master gear used only for checking finished production gears.

(c) Green production control gear. A bevel master gear used only for checking gears in the green or unhardened state.

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6.3.5 Inspected characteristics.

(a) Tooth bearing. That portion of a gear tooth which actually makes contact with its mate.

(b) Top land and back face. As shown in figure 8.

(c) Backlash. The amount by which the width of a tooth space exceeds the thickness of the engaging tooth, as measured by holding one gear fixed, rotating the mating gear, and reading the total movement on a dial indicator positioned normal to the active tooth surface.

(d) Runout. Runout is the total variation in the distance between a surface of revolution and a reference surface, measured perpendicular to the surface of revolution. The surface of revolution of a bevel gear is the pitch cone. The reference surface of a bevel gear is the pitch surface which is contacted by a specified probe as shown in figure 11 and/or figure 12.

(e) Tooth spacing. The distance between corresponding sides of adjacent teeth taken in a plane which is perpendicular to the axis of the gear.

(f) Pitch variation. Pitch variation of bevel and hypoid gears is the variation in spacing between corresponding sides of adjacent teeth, measurements preferably being taken near the middle of the tooth height at the midface.

(g) Maximum index error. Maximum index errors of a gear is the greatest displacement of a tooth with respect to any other tooth measured near the pitch circle.

(h) Required tooth contact. The required tooth contact determination is best accomplished by means of a testing machine. Proper contact involves three factors; namely, position, shape, and area of contact.

6.4 Inspection procedure.

6.4.1 Testing. The inspection procedure involves a progressive proving process known in the industry as "testing." Testing is in addition to all basic inspections, such as for diameters, thicknesses, shoulders, bores, etc. Testing is a functional, rotational check and is

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introduced in the manufacturing process immediately following cutting the teeth in the green blank and continues through all subsequent operations which in any way affect the trueness, shape or size of the finished gear. "Testing" therefore is an essential phase of bevel gear inspection.

6.5 Testing bevel gears.

6.5.1 Inaccuracies. Smooth, quiet-running bevel gears must have correct tooth shapes, uniform spacing of teeth, and teeth concentric with the bore or shank. Inaccuracies in any of these conditions produces noise and rapid wear on high-speed or heavily-loaded installations, and non-uniform motion in the case of precision instruments and control equipment. Testing machines provide a practical method for determining that the gears are right before assembly, and for precisely controlling the quality of the gears during the stages of their manufacture.

6.5.2 Running test. In the testing machine, gears are mounted in the operating position and run under power and light brake load, simulating operation under normal running conditions in their final mountings. Smoothness and quietness of operation, tooth bearing, tooth size, surface finish, runout, and tooth spacing errors are all checkad by the running test.

6.5.3 Tooth bearing. Tooth bearing is an indication of correct tooth shape both up and down and lengthwise on the tooth profile. It is that portion of the gear tooth surface which makes contact with its mate and can readily be observed by painting the teeth with a marking compound and running the gears for a few seconds in a bevel gear testing machine.

6.5.4 Bevel gear testing machines. Bevel gear testing machines are provided with all necessary adjustments to bring the axes of the gears into a relationship that optimum tooth bearing is obtained. By means of these adjustments, it is possible to determine how the gears will operate in and out of position in their final mountings and to measure directly the slight adjustments which must be made in the cutting or grinding machine to locate the tooth bearing correctly. To insure resetting of the testing machines to duplicate the original proven settings for a particular pair of bevel gears, special set-up gages are employed. The use of set-up gages is a practical solution to obtain, on repeat orders, the same quality as originally produced.

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6.6 Testing practice.

6.6.1 Trial or development pair. The routine for manufacture of bevel gears consists of first preparing a "trial or development pair" to be ultimately inspected and approved for operation in the final mounting. Following approval, several sets of master control gears are made which duplicate the "developed pair." These hardened control gears, tailored to satisfy requirements at various operations as production progresses, are then used in the testing machines to control quality at all stages of manufacture. Production gears are tested to the control pinion and the pinions tested to the control gears. This method insures positive control of tooth bearing and makes possible manufacture of quality bevel gears on a production basis.

6.7 Bevel gear measuring instruments.

6.7.1 Inspection devices. On certain high precision gear work, such as for instruments and for aircraft, extremely close tolerances of tooth spacing and concentricity must be held. Checking of these values by sight and sound on a testing machine is not sufficient; they must also be inspected by direct measurement. Individual elemental inspections must be made to prove setups, cutter action, machine performance, tooling, etc. For these purposes three inspection devices are employed; the composite error tester, the tooth spacing tester, and the concentricity tester.

6.7.2 Composite error test. The composite error test is a gear rolling operation for determining size variations, runout, trueness errors, tooth spacing errors, tooth bearing, and backlash.

6.7.3 Trueness errors. Trueness errors are represented as slight oscillations in the indicator as each tooth goes in and out of mesh during rotation in close engagement without load. These oscillations may or may not be uniform in magnitude and are caused by the modification in the tooth profile shape (necessary to localize the tooth bearings) and by tooth spacing errors. The oscillation caused by modification in the tooth profile shape is referred to as the profile wave. This will be approximately equal to the average value of all the individual oscillations around the gear.

6.7.4 Spacing errors or tooth defects. Thick or thin teeth or the presence of dirt or chips are indicated by sudden jumps of the indicator either up or down, on top of, or in addition to the profile wave.

6.7.5 Runout. Runout is determined by rolling the gear with a mating member or control gear known to have no appreciable runout. Runout shows up as a gradual increase in the indicator reading up to a

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high point and a gradual decrease in the indicator reading to a low point, which is generally 180° around the circumference of the member being tested from the high point. Since the amount of the profile wave is the same for every tooth, it has nothing to do with runout. Therefore, it must be subtracted from the difference between the high and low reading to obtain the actual runout. Faulty arbors or improper set-up may show up as a runout in otherwise good gears.

6.7.6 Tooth spacing tester. This instrument measures the variation in spacing from tooth to tooth. Two contact points are employed, one a fixed stop, the other movable to actuate a dial indicator. In use, each tooth of the gear is brought to bear against the fixed stop with the movable point contacting the corresponding flank of the neighboring tooth. The dial indicator reading is noted and the operation is then repeated for all the teeth in the gear.

6.7.7 Concentricity tester. Concentricity of the pitch circle with the center about which the tester turns is measured by the concentricity tester. The runout is twice the eccentricity. This instrument employs two contact points 180° apart, one a fixed stop, the other actuating a dial indicator. The gear to be checked is mounted on a dead center held in the vise, and indicator readings are taken on diametrically opposite teeth with the contacts acting on corresponding sides of these teeth. The contact points are lifted while the gear is turned from tooth to tooth. The runout is equal to one-half the full indicator movement (FIM), which make this method of inspection extremely accurate.

NOTE: Because of the great amount of time required in inspecting gears by these last two devices, they are used primarily for periodic and not general production testing.

6.7.8 Special checking fixtures. Testing machines are preferred over special fixtures because they show how the gears will operate in service. They provide adjustments necessary for determining slight changes in set-up of the cutting machine for control of tooth bearing.

6.8 Checking of tooth profiles.

6.8.1 Tooth profiles. The checking of the tooth profiles is not carried out on bevel gears for two principal reasons. First, the shape of a bevel gear tooth varies at every position from end to end, instead of being constant as in a spur gear. Hence, a different reading is obtained at every checking position along the gear tooth and even a slight variation in checking position from gear to gear will give incorrect results. Second, even if this check were practical for bevel

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gears, it would not show the true contact along the entire length of the tooth, as shown by a running test in a testing machine.

TABLE III. Readings in ten-thousandths inch.

Tooth pair	Indicator reading	Difference between successive readings
1	0	
		+2
2	+2	
		+1
3	+3	
		+1
4	+4	
		+1
5	+2	
		-2
6	0	
		-2
7	+1	
		+1
8	-2	
		-3
9	0	
		+2
10	+3	
		+3
1	0	
		-3
		SUM 0

Since the gear has a full number of teeth, spacings which are too large must be offset by others which are too small. Therefore, the sum of all the differences (column 3) must be zero. If uncompensated errors are made in reading the indicator, the sum of column 3 will not equal zero. If the sum of column 3 deviates from zero by more than .0002 inch, the check must be repeated. The largest number (regardless of sign) in column 3 is the maximum tooth to tooth spacing error.

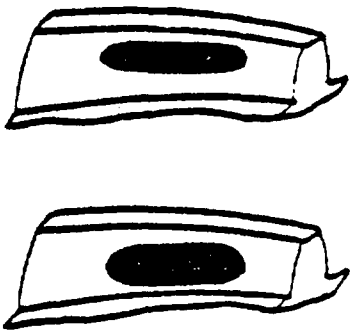
6.9 Superseding. This document supersedes AVCO, Lycoming Division specification P7800C, dated 27 September 1979.

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Note: Also a typical optimum bearing for coniflex straight, and zerol gear sets.

FIGURE 1. A typical optimum tooth bearing for a spiral bevel gear set

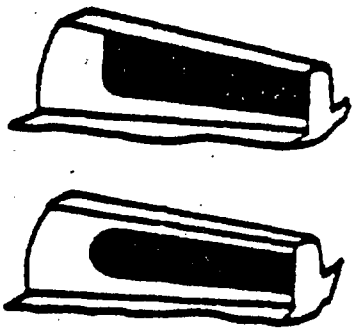
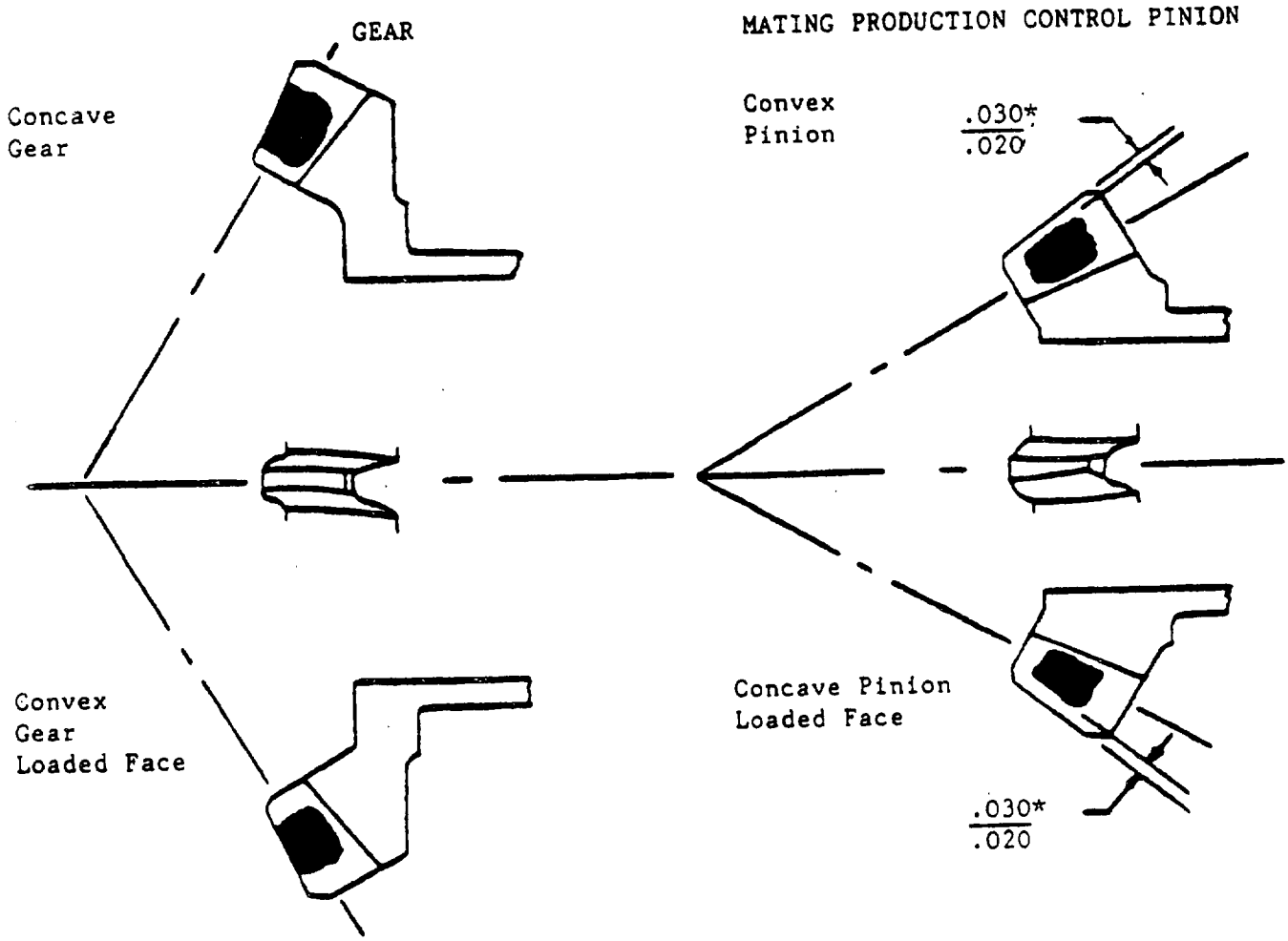


FIGURE 2. Optimum tooth bearing for straight bevel gear not of the coniflex type

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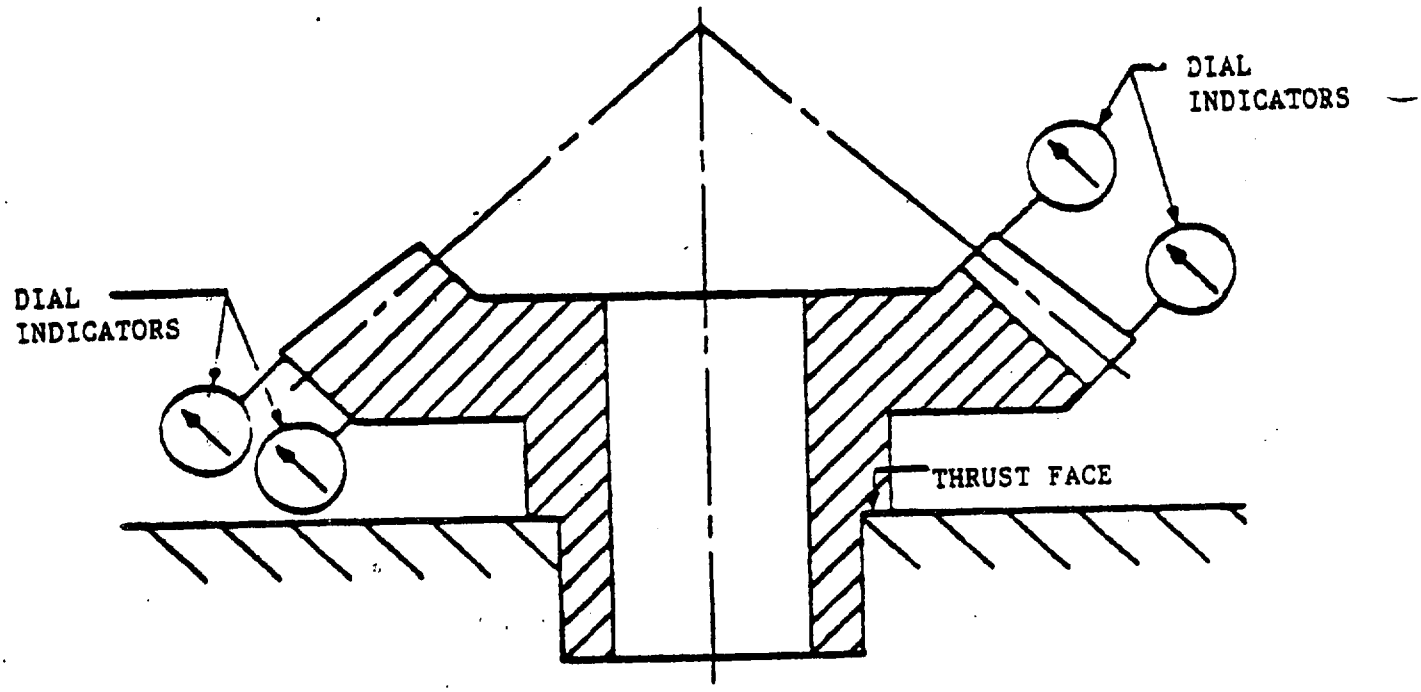


Scale ≈ 2X size
*Full size dimensions

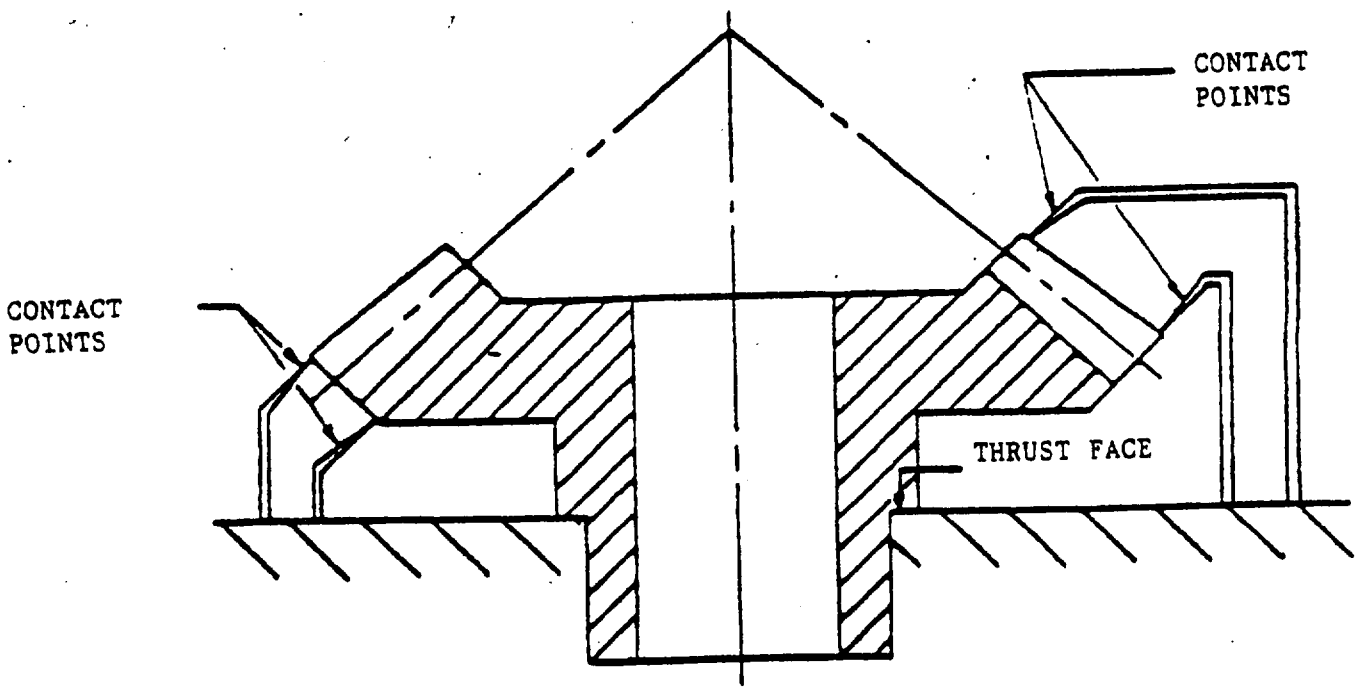
Note: In addition to meeting the pattern diagram, the tooth bearing pattern shall also show the .020 to .030 inch band at the top land edge of the mating basic master pinion at the normal operating distance.

FIGURE 3

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PART A



PART B

FIGURE 4. Two methods for checking the location of top land and back face

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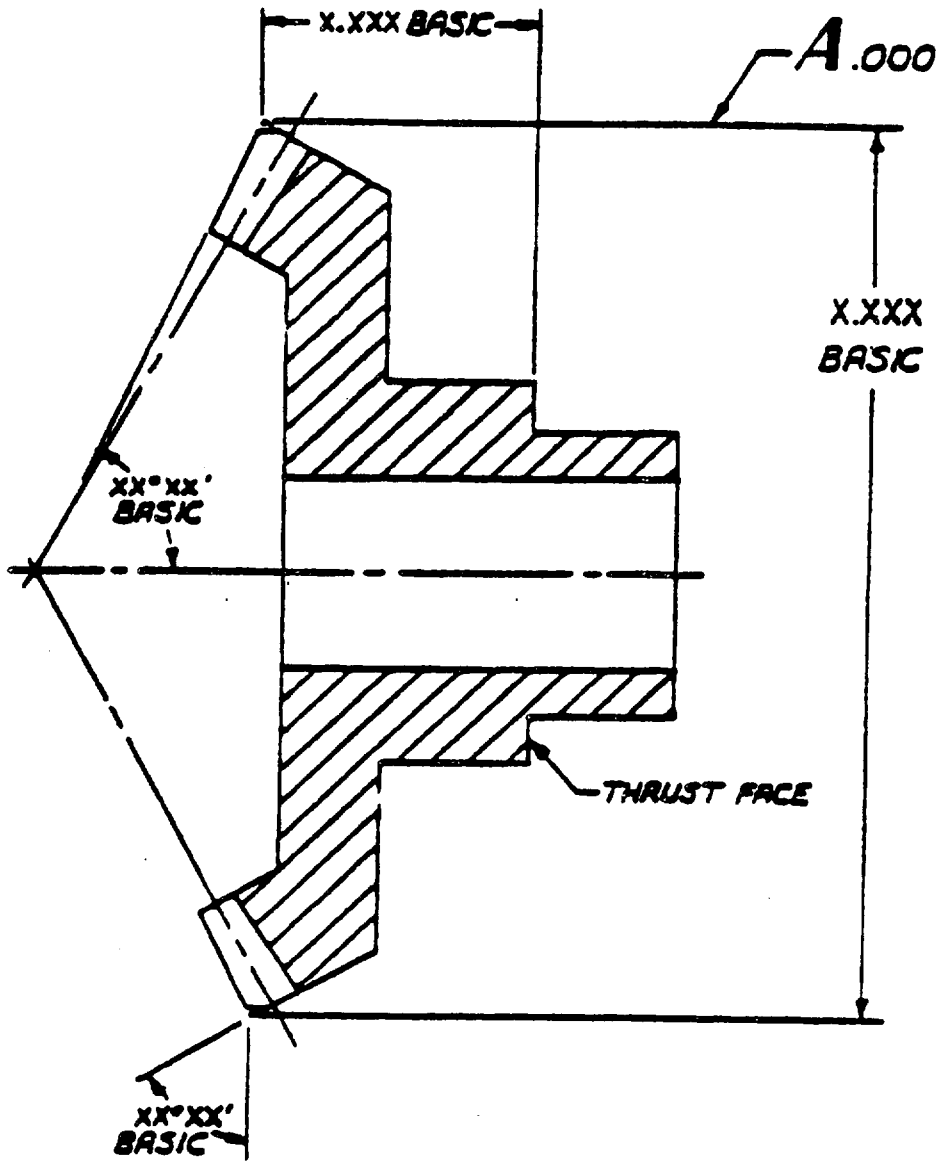


FIGURE 5. Basic dimensions furnished on each bevel gear drawing

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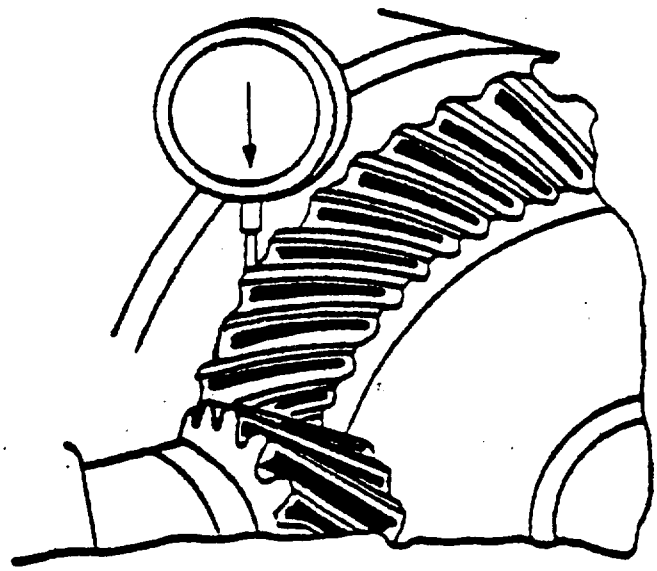
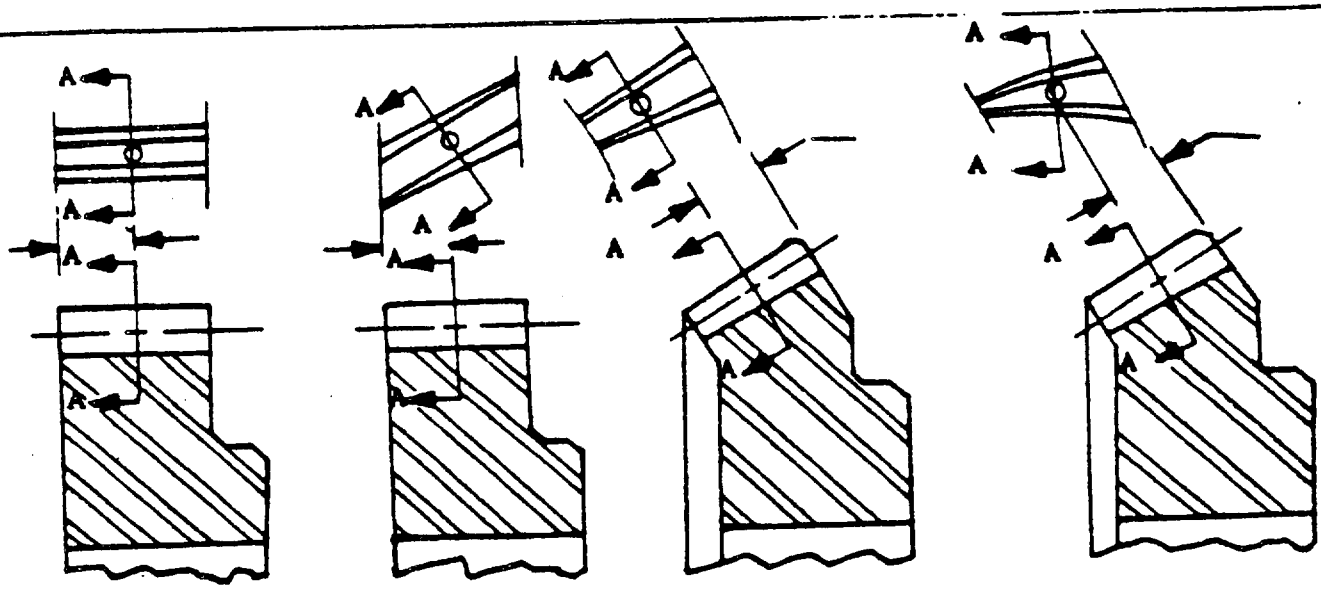
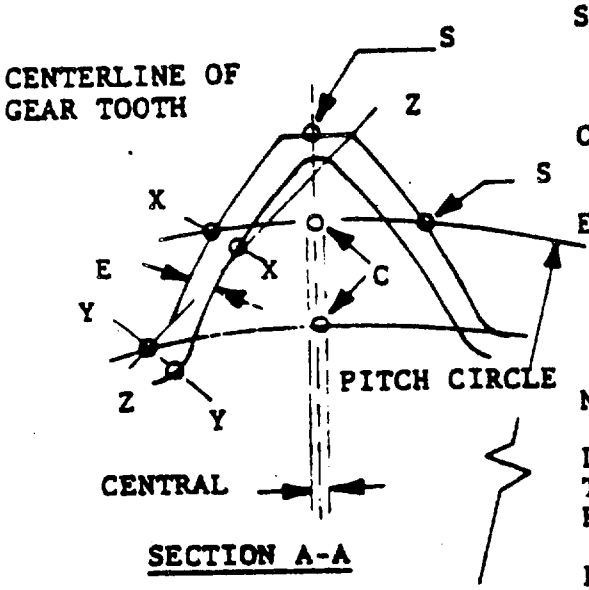


FIGURE 6. Backlash measurement

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( ) SPUR GEAR ( ) HELICAL GEAR CUT NORMAL TO HELIX ANGLE ( ) STRAIGHT BEVEL GEAR ( ) SPIRAL BEVEL GEAR CUT NORMAL TO SPIRAL ANGLE ( ) HYPOID GEAR CUT NORMAL TO SPIRAL ANGLE



SECTION A-A

S = SURFACE HARDNESS _____
( ) TOP LAND OF TOOTH
( ) NORMAL TO TOOTH SURFACE AT PITCH CIRCLE

C = CORE HARDNESS _____
( ) AT PITCH CIRCLE
( ) AT ROOT CIRCLE

E = EFFECTIVE CASE DEPTH _____
CASE HARDNESS _____
( ) TRAVERSE ALONG LINE X-X NORMAL TO TOOTH SURFACE
( ) TRAVERSE ALONG LINE Y-Y PERPENDICULAR TO LINE Z-Z

NOTES:
1. EFFECTIVE CASE DEPTH IS THE LENGTH OF LINE X-X OR LINE Y-Y MEASURED FROM THE SURFAC TO THE LAST POINT TOWARD THE CORE HAVING THE REQUIRED MINIMUM HARDNESS OF ROCKWELL C ____.
2. LINE Z-Z IS TANGENT TO ROOT FILLET AN PASSING THROUGH OPPOSITE TIP CORNER.

LEGEND: (X) APPLICABLE (-) NOT APPLICABLE

FIGURE 7. HARDNESS DETERMINATION METHODS

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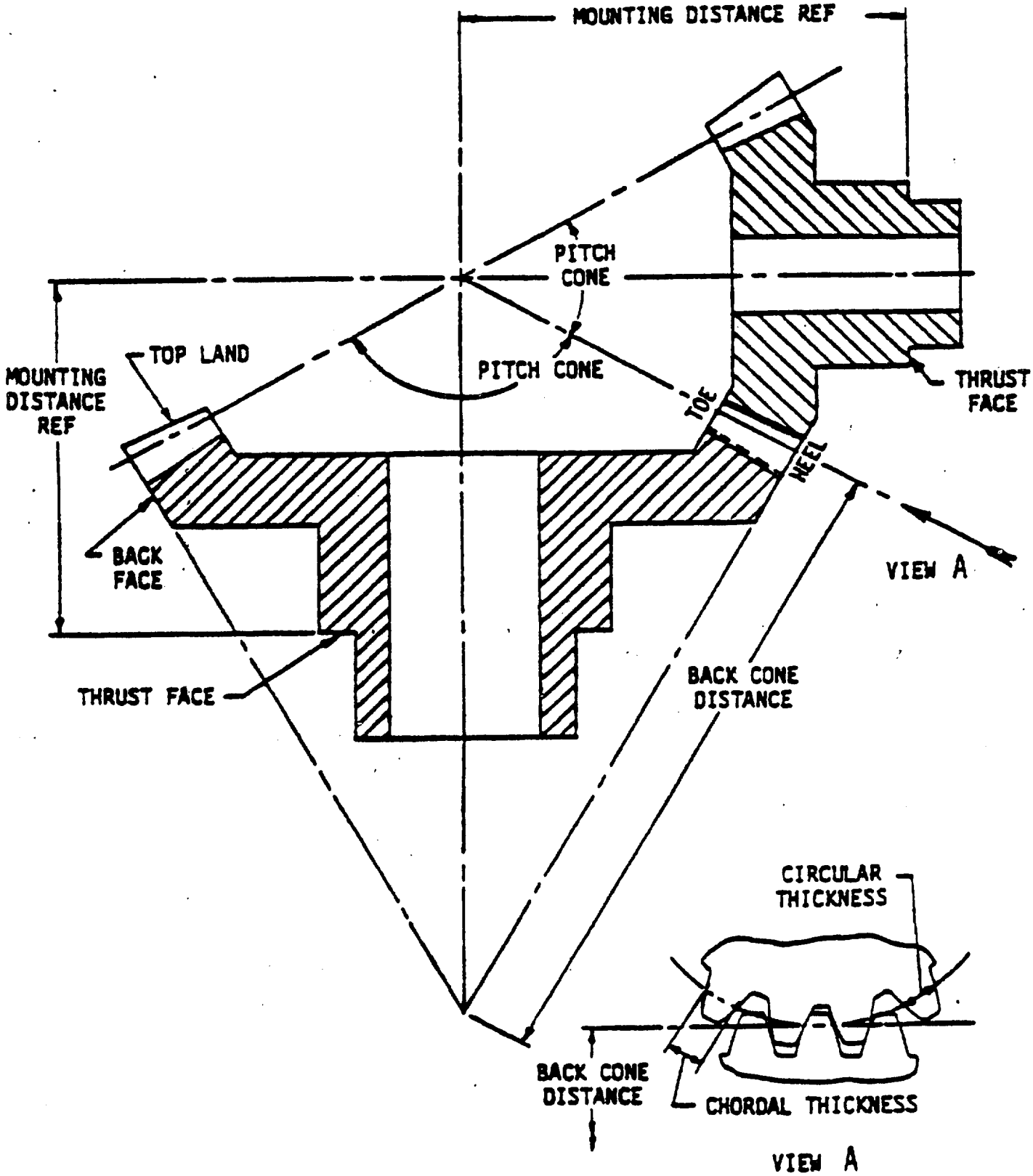


FIGURE 8. Bevel gear nomenclature

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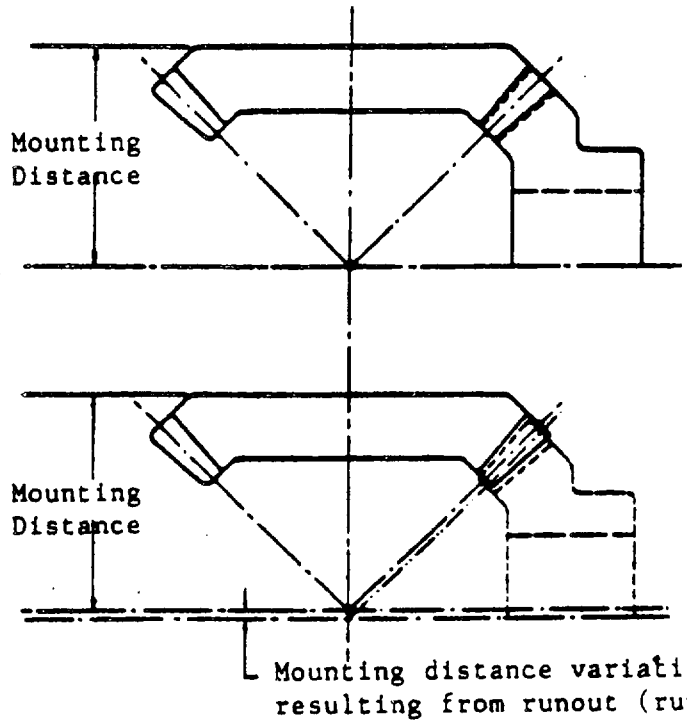
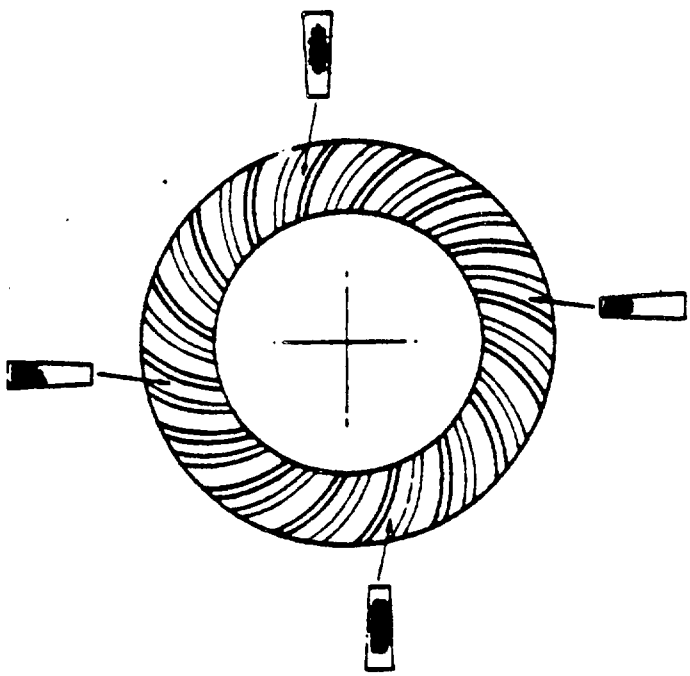


FIGURE 9. Rolling check



Shifting of tooth contact shows presence of runout. Sound variation also characterizes the existence of runout.

FIGURE 10. Runout bearing variation

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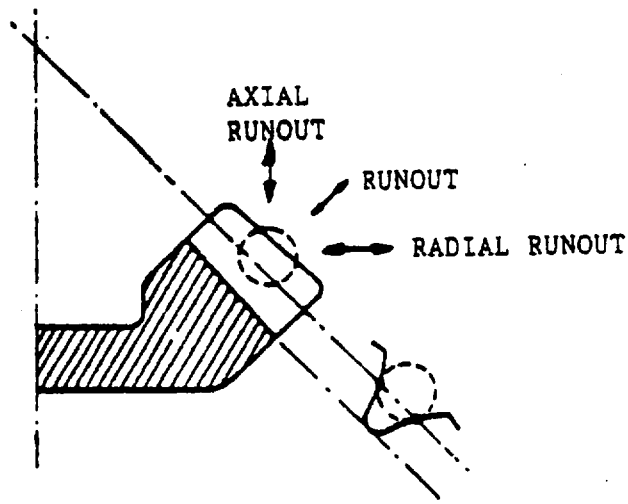


FIGURE 11. Single probe runout check

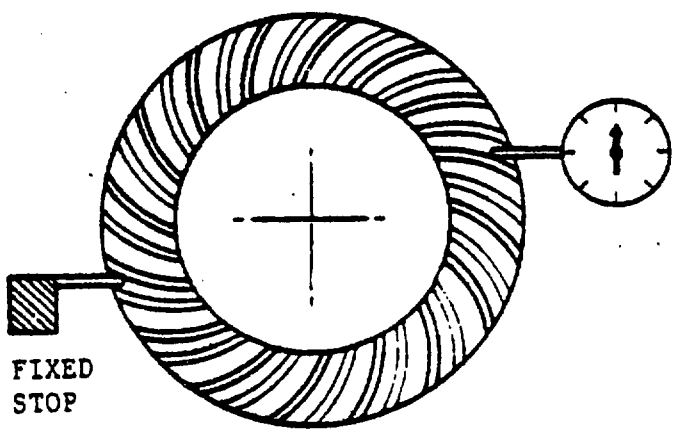


FIGURE 12. Two probe runout check

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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